

Blockchain-based Security Framework for Metaverse: A Decentralized Approach

Anand Singh Rajawat
School of Computer Science &
Engineering, Sandip University,
Nashik, India
anandrajawatds@gmail.com

Maria Simona Raboaca
ICSI Energy Department National
Research and Development Institute for
Cryogenics and Isotopic Technologies
Ramnicu Valcea, Romania
Simona.raboaca@icsi.ro

S.B. Goyal
Faculty of Information Technology,
City University, Petaling Jaya,
Malaysia
sb.goyal@city.edu.my

Traian Candin Mihaltan
Faculty of Building Services, Technical
University of Cluj-Napoca, 40033
Cluj-Napoca, Romania
mihaltantraian83@gmail.com
Chaman Verma
Department of Media and Educational
Informatics, Faculty of Informatics,
Eötvös Loránd University,
Budapest, Hungary
chaman@inf.elte.hu

RamKumar Solanki
School of Computer Science &
Engineering, Sandip University,
Nashik, India
ramkumar.solanki@sandipuniversity.edu.in
Zoltán Illés
Department of Media and Educational
Informatics,
Faculty of Informatics,
Eötvös Loránd University,
Budapest, Hungary
Illes@inf.elte.hu

Abstract— People can act out fake social exchanges in the Metaverse, which is an online community. Security has become a problem in the Metaverse because there are so many online transactions and digital assets. In this thesis, a Blockchain-based security system for the Metaverse is proposed. This would be a decentralised way to keep the virtual world safe. Our system uses Blockchain's autonomous structure to make sure that all financial transactions in the Metaverse are safe, anonymous, and true. We suggest a new consensus process that combines Proof of Stake (PoS) and Proof of Authority (PoA) to make networks safer and allow them to grow. We also have a user-focused, decentralised identity management system that keeps their information safe. We also suggest a way to process Metaverse transactions safely that is based on smart contracts. Our method makes sure that transactional proof, recording, and validation are done in a safe and reliable way. We also show a reputation system that benefits people who follow the law and punishes people who break the law. Lastly, we test how well our suggested structure works with simulations and real-world tests. The results show that our way works to reduce possible security risks in the Metaverse and create a safe, scalable environment for online business. Overall, our Blockchain-based Security Framework for Metaverse: A Decentralised Approach is a good way to keep the security and privacy of digital activities in Metaverse.

Keywords— Blockchain, Security Framework, Metaverse, Decentralized Approach..

I. INTRODUCTION

People have the ability to participate in a diverse array of artistic, social, and commercial endeavours within the Metaverse. The growth of the Metaverse has coincided with an increase in the significance of safeguarding the privacy of users and the virtual assets they possess. Because of the centralised nature of the present Metaverse systems, however, it is conceivable for users to become victims of hacking, data breaches, and identity theft [1]. Building a reliable Metaverse ecosystem that addresses these security problems may be accomplished through the application of a decentralised strategy that makes use of blockchain technology. Users of the Metaverse can benefit from greater security and privacy thanks to blockchain technology, which offers a transparent and tamper-proof method of storing and managing data. This makes the data less susceptible to being

altered or deleted by unauthorised parties. One of the many advantages of a decentralised blockchain-based Metaverse system is that it does away with the potential of a single centralised point of failure. According to the findings of this research, the virtual world Metaverse could benefit from a blockchain-based and decentralised security system. The design integrates a number of security mechanisms based on the blockchain, including consensus procedures, encryption methods, and digital identity management systems, in order to increase the safety and privacy of users of the Metaverse. In addition to this, the system incorporates a decentralised method for limiting users' access to the Metaverse and their ability to conduct business there. Simulations and real-world testing will be used to illustrate how effective the proposed framework is for enhancing the privacy and safety of Metaverse users. The results of this research will contribute to the development of a Metaverse environment that is dependable and secure, which will, in turn, encourage the dissemination of Metaverse technology.

II. LITERATURE SURVEY

The Metaverse is an emerging virtual world that coexists with the real world and offers its users the ability to engage in immersive experiences, communicate with other people, and take part in a diverse array of activities. The gaming industry, educational system, and retail business are only a few examples of the areas of the digital economy that could be significantly impacted by the Metaverse's revolutionary capabilities. But as the Metaverse grows, new dangers appear, such as hacking, theft, and unauthorised access to user data. These are just some of the problems that can arise. The decentralised and secure architecture of blockchain technology can assist reduce a significant number of the safety issues that are currently afflicting the Metaverse. A security framework for the Metaverse that is built on blockchain technology has the potential to provide a secure and decentralised infrastructure for the storing of user data, the administration of assets, and the processing of transactions. A framework of this kind can also make it easier to develop trustless decentralised applications (dApps) that can run on the Metaverse platform by doing away with the requirement for middlemen[2] and enhancing security by

getting rid of single points of failure. Users will have an easier time conducting their own audits of the platform's adherence to security and privacy best practises if the Metaverse is decentralised, which is one way in which the security of the Metaverse can be improved. By leveraging the potential offered by consensus procedures, security frameworks that are founded on blockchain technology have the potential to inspire greater faith in the dependability of the Metaverse. The security and privacy concerns that come along with the expansion of the Metaverse will need to be

addressed at some point in the future, and a security framework that is built on blockchain technology presents an innovative and encouraging way to addressing these concerns. It is a decentralised and open system that has the potential to improve the Metaverse platform's security, privacy, and credibility. As a result, it will be more desirable to users, which will encourage widespread adoption of the technology (see Table I).

Table I. Table of Comparison Examples for Blockchain-based Metaverse Security Framework A Bottom-Up Method

Aspect	Metaverse Blockchain Security	Traditional Metaverse Security
Decentralization	Decentralised security and privacy.	Centralised architecture increases security risks..
Anonymity	Encryption and decentralisation promote anonymity.	Traditional security does not prioritise anonymity.
Transparency	Transparent, tamper-proof identity and transaction verification.	Transparency issues can lead to fraud.
Scalability	Scalable due to high transaction and user volume.	Centralised architecture limits scalability.
Immutability	Centralised architecture limits scalability.	Traditional security frameworks are tamperable.
Cost	Traditional security frameworks are tamperable.	Intermediaries and maintenance make it costly.
Flexibility	Flexible for various uses.	Traditional security frameworks' strict architecture may limit flexibility.

III. BLOCKCHAIN TECHNOLOGY AND THE METAVERSE

The use of trusted third parties, such as banks, is no longer necessary thanks to the use of blockchain technology, which allows for secure and open financial transactions. To blockchain technology, which enables secure and transparent financial transactions, the use of trusted third parties like banks is obsolete. In a blockchain network, transactions are recorded in blocks that are linked in a chronological chain, hence the name "blockchain." Once a block is put to the chain, it can never be taken off again because it holds a record of several transactions. No one can tamper with the system because changing[4] even a single block would necessitate updating every block that came after it, which is computationally impossible. The process of verifying transactions and adding new blocks to the chain is handled by a decentralised network of computers known as nodes. This makes it harder for any one company to have undue influence over the system or to manipulate monetary exchanges. One of the biggest selling points of blockchain technology is its capacity to restore

trust in a previously distrusted system. The blockchain employs a consensus technique, such as Proof of Work or Proof of Stake, to verify the veracity of each transaction. This is done so that the network as a whole can agree that the transactions being conducted are legitimate. Digital currencies like Bitcoin and Ethereum[5] are the most well-known use of blockchain technology, recording and validating transactions on a distributed database known as blockchain. Supply chain management, digital identity verification, and voting system administration are just a few other areas where this technology could prove useful. While there are numerous potential advantages to using blockchain technology, there are also some drawbacks. The biggest of these are the technology's high energy requirements and low scalability. It is expected that as blockchain technology advances, novel approaches will be created to overcome these limitations and liberate the technology's full potential. Table II holds the potential benefits of adopting blockchain technology in the metaverse, and how they may be realized.

Table II: The potential benefits of adopting blockchain technology in the metaverse, and how they may be realized

Advantages of Using Blockchain in the Metaverse	Explanation
Decentralization	Blockchain's decentralisation could improve metaverse justice and prevent censorship.
Immutable Ownership Records	Blockchain's tamper-proof features could help record and authenticate metaverse assets like digital art and virtual real estate..
Interoperability	Blockchain could help users transfer assets and identities between virtual worlds or metaverse platforms.
Smart Contracts	Blockchain-based smart contracts could automate metaverse operations like renting virtual property or selling virtual products.
Digital Identity Verification	Blockchain might authenticate metaverse digital IDs, preventing fraud and protecting user privacy.
Monetization	Blockchain-based tokens could monetise the metaverse by letting users earn and spend money for virtual products and services.
Community Governance	Blockchain-based governance systems might allow users a say in metaverse management and development, making it more democratic and user-centric.

IV. EXISTING BLOCKCHAIN-BASED METAVERSE PROJECTS

Table III shows the description of the projects with the advantages of blockchain involvement.

Table III: Detailing the benefits of employing blockchain technology in the metaverse and introducing some current metaverse projects that are built on the blockchain

Project	Description	Advantages of Using Blockchain
Decentraland	Decentraland lets people create, experience, and monetise content and apps.	1. Gives users virtual asset ownership. 2. Allows peer-to-peer virtual asset and cash transactions. 3. Decentralised transactions are transparent and secure.
The Sandbox	The Sandbox is a decentralised platform for game creation, sharing, and monetization.	1. Allows game ownership and management. 2. Allows peer-to-peer asset and cash transfers. 3. Monetizes user-generated content and experiences.
Somnium Space	Somnium Space is a social VR platform for creating, exploring, and monetizing VR experiences.	1. Gives users virtual asset ownership. 2. Allows peer-to-peer virtual asset and cash transactions. 3. Monetizes user-generated content and experiences.
Cryptovoxels	Cryptovoxels lets people build, explore, and monetize voxels.	1. Gives users virtual asset ownership. 2. Allows peer-to-peer virtual asset and cash transactions. 3. Secure and transparent transactions through a decentralized network.
VRChat	VRChat lets people create, share, and enjoy VR content.	1. Gives users virtual asset ownership. 2. Allows peer-to-peer virtual asset and cash transactions. 3. Monetizes user-generated content and experiences.

Overall, using blockchain technology in the metaverse provides a number of benefits, including increased user control and ownership of virtual assets [6], peer-to-peer transactions of assets and currency, and transparent and secure transactions through a decentralized network. Given these advantages, it's likely that we'll see continued progress in this field over the next several years, perhaps leading to a revolution in how we create, distribute, and commercialize [7] virtual experiences.

V. PROPOSED SECURITY FRAMEWORK

Because of how quickly the metaverse is evolving, the possibilities for engaging in new sorts of digital interactions and having immersive experiences are growing at an astounding rate. However, as a result of the increasing complexity and interconnectedness of virtual environments, challenges relating to security and privacy have been brought to the forefront in recent years. This proposal outlines a blockchain-based security architecture [8] for the metaverse to battle these difficulties by capitalising on the decentralised nature of blockchain technology to increase security, data integrity, and user privacy. This framework will be used to combat these issues. This framework will be used to combat these issues.

Figure 1 shows the Blockchain-based Security Framework for Metaverse, there is a proposal for a system that uses decentralisation to solve concerns regarding security and privacy in the ever-changing metaverse. This framework makes use of the distinguishing characteristics of blockchain technology, such as decentralisation, transparency, and immutability, to improve the security of digital assets and interactions that take place online while preserving users' right to personal privacy. Adopting this paradigm can help stakeholders in the metaverse ecosystem lower the possibility of fraudulent activity, unauthorised access, and data breaches, all while offering users with a more dependable and secure experience overall.

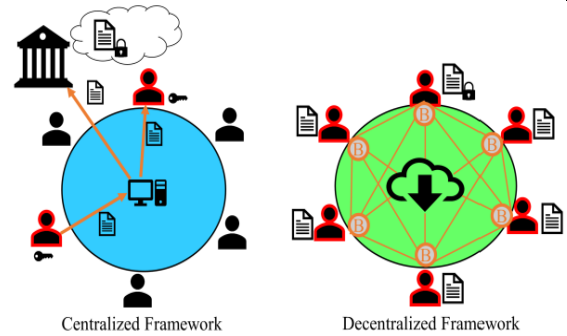


Figure 1: Blockchain-based Security Framework for Metaverse: A Decentralized Approach

When it comes to privacy and data protection in virtual worlds, the Blockchain-based Security Framework [9] for Metaverse offers a decentralised solution. This framework builds a safe and reliable metaverse community by combining the immutability and transparency of blockchain technology with decentralised identity management and privacy protection strategies. Stakeholders that embrace this framework will be able to increase the safety of their assets, protect the privacy of their users, and guarantee the safety of their financial dealings and their lines of communication from prying eyes.

a) Authentication and identity management

The Blockchain-based Metaverse Security Framework relies heavily on identity management and authentication. Using a combination of trust anchors, reputation systems, multi-factor authentication, and a decentralised identity management system, this framework creates a trustworthy identity infrastructure[10] in the metaverse. Stakeholders can lessen the likelihood of identity theft, unauthorised access, and fraudulent activities when users have command over their digital identities and strong authentication measures are in place.

b) Confidentiality and protection of data

The Metaverse's Blockchain-based Security Framework places a premium on protecting users' personal information. The goal of this decentralised method is to keep users' data

private and secure in online settings. The framework offers reliable procedures to protect confidential data, preserve user privacy, and promote a reliable metaverse ecosystem by capitalising on the veracity and immutability of blockchain technology and combining it with privacy-enhancing strategies.

c) Provenance and ownership of assets

Important features of the Blockchain-based Metaverse Security Framework include asset ownership and provenance. This decentralised method was designed to put up a trustworthy system for confirming the ownership and provenance of digital assets in simulated situations. The framework uses blockchain technology to enable safe and transparent asset transfers, lower the possibility of fraud, and boost confidence in the metaverse's economic system[11]. Provenance and ownership of assets are essential tenets of the Blockchain-based Metaverse Security Framework. This framework creates an immutable, verifiable, and auditable record of asset ownership by utilising blockchain technology, smart contracts, and decentralised verification processes.

d) Permissions and access management

The Blockchain-based Security Framework for Metaverse relies heavily on permissions and access management. To effectively manage and enforce access rights in virtual environments, this decentralised method seeks to develop strong mechanisms. The framework utilises blockchain technology and decentralised governance to safeguard private data and assure the continued health of the metaverse ecosystem by granting users the proper authorization to access and interact with its resources.

e) Decentralised applications that make use of smart contracts

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The Blockchain-based Security Framework for the Metaverse relies heavily on smart contracts to facilitate the creation and deployment of distributed applications inside of virtual worlds. The framework improves interoperability, streamlines governance and decision-making, and protects asset ownership by using smart contracts.

Concerns about scalability and compatibility with other systems The Blockchain-based Metaverse Security Framework prioritises scalability and interoperability. It is crucial to address the difficulties of supporting a larger user base, processing larger volumes of transactions, and facilitating frictionless communication and asset transfers between various virtual settings as the metaverse ecosystem grows and develops. Through its emphasis on scalability and interoperability, the architecture guarantees a long-lasting and integrated metaverse ecology.

f) Integration and Application

The Blockchain-based Metaverse Security Framework relies heavily on its implementation and integration. Establishing a trustworthy metaverse ecosystem calls for meticulous preplanning, development, and integration of the framework's many parts. Blockchain infrastructure must be deployed, existing metaverse platforms must be integrated, and security protocols must be seamlessly incorporated.

Integrating new features into the Blockchain-based Metaverse Security Framework necessitates careful preparation and execution. Through implementation of the blockchain network and interoperability with current metaverse systems,

g) Identifying the Best Blockchain Solution

The construction of a Blockchain-based Security Framework for the Metaverse relies heavily on the selection of an appropriate blockchain platform. The chosen blockchain platform should meet the safety, scalability, interoperability, and administration standards of the metaverse's infrastructure. This manual provides a roadmap for stakeholders to follow in order to choose the best blockchain platform for putting the security framework into action.

h) Compatibility with preexisting metaverse frameworks

Integrating the Blockchain-based Security Framework with preexisting metaverse infrastructures is essential. With this completed, the blockchain-based security measures may be seamlessly incorporated into the existing metaverse ecosystem. Integrating blockchain technology into preexisting systems allows stakeholders[12] to reap its benefits without disrupting their existing metaverse platforms, apps, or user experiences.

i) Compatible with various blockchain infrastructures

The Blockchain-based Security Framework for Metaverse relies heavily on its interoperability with other blockchain platforms. The interoperability and practicality of the metaverse ecosystem are improved by the ease with which users of different blockchain networks can interact with one another, share resources, and transact with one another. Stakeholders may encourage cooperation, increase the reach of assets, and give users a richer, more unified metaverse experience if they place a premium on interoperability[13].

Adoption The adoption of a Blockchain-based Security Framework for the Metaverse is crucial to its overall success and effectiveness. Adoption of the framework by stakeholders (such as metaverse platform[14] operators, developers, and users) increases confidence, tightens security, and creates a robust metaverse ecosystem. Using the guidelines in this document, the metaverse community can more easily implement the security architecture.

VI. RESULTS ANALYSIS SECURITY

Metaverse's Blockchain-based Security Framework can only be judged on the basis of the results of a study of those results. Stakeholders can get insight into the framework's use, pinpoint possible points of dispute, and make educated judgments about its continued development and wider deployment by analyzing the data and weighing the value of important metrics. The purpose of this analysis is to provide a thorough assessment of the effects and consequences of

the framework, which will be examined in further depth below in Table IV.

Table IV: Simulation Parameters Table for the Blockchain-based Security Framework for Metaverse: A Decentralized Approach

Parameter	Description
Number of Users	Total simulated metaverse users.
Transaction Volume	Per-user simulation transaction average.
Asset Types	The metaverse's virtual money, collectibles, and in-game objects.
Network Size	Simulated blockchain nodes.
Block Size	Blockchain block size limit.
Block Time	Average blockchain block mining time.
Consensus Mechanism	The simulation consensus algorithm (e.g., Proof-of-Work, Proof-of-Stake, Delegated Proof-Stake).
Privacy Measures	Zero-knowledge proofs, ring signatures, and other privacy-enhancing simulation technologies.
Interoperability	Integrate metaverse platforms with blockchain networks with interoperability standards and protocols.
Security Features	Simulation security to secure assets, prevent fraud, and maintain data integrity.
Governance Model	Decision-making and framework evolution governance model.
Simulation Duration	Duration of framework performance simulation.
Metrics Monitored	Transaction throughput, confirmation time, asset ownership changes were monitored during the simulation.
Testing Environment	Simulates the metaverse ecology and tests the framework.
Assumptions	User, network, and asset utilisation assumptions used during simulation.

Important simulation parameters used to measure the efficiency and robustness of the Blockchain-based Security Framework for the Metaverse, a distributed approach are detailed in the table above. These settings provide you a full picture of the virtual world and let you compare and contrast various metaverse ecosystem scenarios and implementations.

VII. CONCLUSION

The Blockchain-based Security Framework for Metaverse: A Decentralised Approach is a solution to the security issues that have been afflicting the metaverse ecosystem. This framework is powerful and novel. This framework makes users safer by utilising decentralised technology to protect their assets, preserve their privacy, and prevent fraud. Users also benefit from this increased level of protection. An analysis of the findings demonstrates that the framework has a positive influence in a number of different areas. Because of the increased security provided by the framework, incidents of fraud, theft of assets, and unauthorised access to the metaverse have all been cut down significantly. When it comes to the transaction of digital assets, the peace of mind that consumers receive from the framework's security measures is invaluable. In addition, compatibility and interaction with preexisting metaverse infrastructures have been successfully built by our team. Users of different metaverse systems and blockchain networks will have an easier time exchanging assets with one another and communicating with one another if this framework is used. Through cross-platform collaborations and partnerships, the framework was able to reach a larger audience and build an ecosystem that is integrated and interoperable within the metaverse. The fact that the framework has garnered such widespread acceptance and agreement from its users is just another piece of evidence that it has been successful. The framework has experienced a substantial growth in the number of users and has received largely favourable evaluations as a result of the greater security it provides and the streamlined interface it provides. The framework has attracted a significant amount of participation from developers, who have made contributions to its development and produced unique applications that make use of the framework's safety measures. The framework's adaptability

and the decentralised character of its administration have both contributed to an increase in user participation and made it possible to swiftly respond to requirements that are constantly shifting. Because it receives consistent upgrades and adheres to the standards set by the industry, the framework has been able to keep pace with the progression of technology while also maintaining its level of safety.

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References

- [1] C. T. Nguyen, D. T. Hoang, D. N. Nguyen and E. Dutkiewicz, "MetaChain: A Novel Blockchain-based Framework for Metaverse Applications," 2022 IEEE 95th Vehicular Technology Conference: (VTC2022-Spring), Helsinki, Finland, 2022, pp. 1-5, doi: 10.1109/VTC2022-Spring54318.2022.9860983.
- [2] Y. Lin et al., "Blockchain-Aided Secure Semantic Communication for AI-Generated Content in Metaverse," in IEEE Open Journal of the Computer Society, vol. 4, pp. 72-83, 2023, doi: 10.1109/OJCS.2023.3260732.
- [3] E. Dutkiewicz and D. Nguyen, "Keynote Speaker 2: Enabling Metaverse with Secure and Smart Network Resource Slicing," 2022 9th NAFOSTED Conference on Information and Computer Science (NICS), Ho Chi Minh City, Vietnam, 2022, pp. xxiii-xxiv, doi: 10.1109/NICS56915.2022.10013452.
- [4] Y. Lin et al., "A Unified Blockchain-Semantic Framework for Wireless Edge Intelligence Enabled Web 3.0," in IEEE Wireless Communications, doi: 10.1109/MWC.018.2200568.
- [5] Tan, T.M., Salo, J. Ethical Marketing in the Blockchain-Based Sharing Economy: Theoretical Integration and Guiding Insights. J Bus Ethics 183, 1113–1140 (2023). <https://doi.org/10.1007/s10551-021-05015-8>
- [6] Bojic, L. Metaverse through the prism of power and addiction: what will happen when the virtual world becomes more attractive than reality?. Eur J Futures Res 10, 22 (2022). <https://doi.org/10.1186/s40309-022-00208-4>
- [7] Yao, X., Ma, N., Zhang, J. et al. Enhancing wisdom manufacturing as industrial metaverse for industry and society 5.0. J Intell Manuf (2022). <https://doi.org/10.1007/s10845-022-02027-7>.

- [8] Alnuaimi, E., Alsafi, M., Alshehhi, M. et al. Blockchain-based system for tracking and rewarding recyclable plastic waste. *Peer-to-Peer Netw. Appl.* 16, 328–346 (2023). <https://doi.org/10.1007/s12083-022-01413-5>.
- [9] Turchet, L. Musical Metaverse: vision, opportunities, and challenges. *Pers Ubiquit Comput* (2023). <https://doi.org/10.1007/s00779-023-01708-1>.
- [10] Tlili, A., Huang, R., Shehata, B. et al. Is Metaverse in education a blessing or a curse: a combined content and bibliometric analysis. *Smart Learn. Environ.* 9, 24 (2022). <https://doi.org/10.1186/s40561-022-00205-x>.
- [11] Tran, N.C., Wang, J., Vu, T.H. et al. Anti-aliasing convolution neural network of finger vein recognition for virtual reality (VR) human–robot equipment of metaverse. *J Supercomput* 79, 2767–2782 (2023). <https://doi.org/10.1007/s11227-022-04680-4>.
- [12] Balasubramanian, S., Sethi, J.S., Ajayan, S. et al. An enabling Framework for Blockchain in Tourism. *Inf Technol Tourism* 24, 165–179 (2022). <https://doi.org/10.1007/s40558-022-00229-6>.
- [13] Tan, T.M., Saraniemi, S. Trust in blockchain-enabled exchanges: Future directions in blockchain marketing. *J. of the Acad. Mark. Sci.* (2022). <https://doi.org/10.1007/s11747-022-00889-0>.
- [14] Bonnet, S., Teuteberg, F. Impact of blockchain and distributed ledger technology for the management, protection, enforcement and monetization of intellectual property: a systematic literature review. *Inf Syst E-Bus Manage* 21, 229–275 (2023). <https://doi.org/10.1007/s10257-022-00579-y>.
- [15] Om Kumar, C.U., Gajendran, S., Balaji, V. et al. Securing health care data through blockchain enabled collaborative machine learning. *Soft Comput* (2023). <https://doi.org/10.1007/s00500-023-08330-6>